AX-1177 — Methods Annex

Programme: AX-1177 (anti-IL-33 monoclonal antibody) **Document type:** Methods annex and dataset annotation **Applies to:** in vitro transcriptomics dataset; clinical serum proteomics dataset **Status:** Final

This annex records materials, experimental design, instrument settings and analysis parameters. It records no results and no interpretation.

1. Test and control articles

Article	Description	Format	Working concentration
Compound_47	Anti-IL-33 monoclonal antibody, binds the reduced and oxidised forms of human IL-33	Human IgG1	10 µg/mL
NIP228	Non-binding isotype control antibody; no known human antigen	Human IgG1	10 µg/mL
IL-33red	Recombinant human IL-33, reduced form, mature processed species (residues 112–270)	Recombinant protein	30 ng/mL

Compound_47 is the research designation for AX-1177. NIP228 is the isotype control.

Compound_47 nominal molecular weight 146 kDa. NIP228 nominal molecular weight 146 kDa. IL-33red nominal molecular weight 18.1 kDa.

Vehicle articles: PBS and DMSO.

Affinity, potency and binding characterisation for Compound_47 are held in the programme biologics file and are not part of this dataset.

2. Cell lines

Label	Cell type	Vascular bed	Source	Donor	Passage used
HUVEC	Human umbilical vein endothelial cells	Macrovascular, venous	Commercial primary	Single donor	P2
HGMEC	Human glomerular microvascular endothelial cells	Microvascular	Commercial primary	Single donor	P5
A549	Human lung adenocarcinoma epithelial line	n/a	ATCC CCL-185	n/a	—

MCF7	Human breast adenocarcinoma epithelial line	n/a	ATCC HTB-22	n/a	—

Endothelial lines were maintained in endothelial growth medium with supplements and 2% FBS. A549 and MCF7 were maintained in DMEM with 10% FBS. All lines were confirmed mycoplasma negative prior to plating. Cells were plated to 80–90% confluence and serum-reduced for 4 h before treatment.

3. Experimental design

Cells were treated for 6 h or 24 h. All arms were plated and harvested in parallel within each cell line and timepoint.

Arms, endothelial lines:

Arm	IL-33red	Antibody	Lines
Vehicle	—	—	HUVEC
NIP228	—	NIP228	HUVEC, HGMEC
Compound_47	—	Compound_47	HUVEC, HGMEC
IL-33red	30 ng/mL	—	HUVEC, HGMEC
IL-33red + NIP228	30 ng/mL	NIP228	HUVEC, HGMEC
IL-33red + Compound_47	30 ng/mL	Compound_47	HUVEC, HGMEC

Arms, epithelial lines (A549, MCF7):

Arm	IL-33red	Antibody
PBS	—	—
DMSO	—	—
NIP228	—	NIP228
Compound_47	—	Compound_47

Replication: n = 3 biological replicates per arm per timepoint.

Total samples: 38 arm × timepoint groups, 114 libraries.

Antibody, where present, was added concurrently with IL-33red. No pre-incubation step was used.

4. RNA extraction and sequencing

Total RNA was extracted using column-based purification with on-column DNase treatment. RNA integrity was assessed on an Agilent TapeStation; all samples carried forward had RIN ≥ 8.0. Libraries were prepared with ribosomal RNA depletion. Sequencing was performed on an Illumina NovaSeq 6000, paired-end 2 × 100 bp, targeting 25 M read pairs per library.

All 114 libraries were prepared in a single batch and sequenced across one flow cell.

5. Analysis parameters

Reads were aligned to GRCh38 and summarised to Ensembl gene identifiers. Low-expression genes were removed. Counts were normalised and transformed with voom.

A single linear model was fitted across the full 114-library dataset. The design matrix encoded cell line, timepoint and treatment arm, with one coefficient per arm × timepoint × cell line group. Empirical Bayes moderation was applied across the whole fit. All reported contrasts were extracted from this single fitted model.

Column definitions in the delivered tables:

Column	Definition
`Comparison`	the contrast extracted from the fitted model
`ensembl`	Ensembl gene identifier
`logFC`	log2 fold change for the named contrast
`AveExpr`	average log2 expression for the gene, computed across all 114 libraries in the fit
`t`	moderated t statistic
`P.Value`	unadjusted p value
`adj.P.Val`	Benjamini–Hochberg adjusted p value, computed within contrast
`B`	log-odds of differential expression
`symbol`, `entrez`, `gene_name`	gene annotation

Count matrices and the fitted model object are retained in the programme archive and are not part of this delivery.

6. Clinical serum proteomics

Cohort: 100 patients hospitalised with PCR-confirmed SARS-CoV-2 infection, single-centre, pre-vaccination period. Serum collected by venesection at enrolment, clotted, separated by centrifugation, stored at –80 °C.

Composite endpoint: admission to intensive care, requirement for invasive mechanical ventilation, or death within 30 days of hospitalisation. Recorded as a binary variable.

Assay: Olink Target 96 proximity extension assay. Panels run:

Inflammation (v.3022), Cardiovascular II (v.5006), Cardiovascular III (v.6114), Cardiometabolic (v.3603), Cell Regulation (v.3703), Development (v.3521), Immune Response (v.3203), Metabolism (v.3403), Neuro Exploratory (v.3911), Neurology (v.8013), Oncology II (v.7004), Oncology III (v.4001), Organ Damage (v.3311).

Each panel reports its own assay set. Assay identifiers are panel-specific and are reported as delivered by the vendor without renaming or reconciliation across panels.

Model: logistic regression of the composite endpoint on each assay, one assay per model. Biomarker values were log transformed. Models were adjusted for age, sex and National Early Warning Score 2 at enrolment. Odds ratios are reported per log unit with 95% confidence intervals. Benjamini–Hochberg adjustment was applied across all assays in the delivery.

7. Post-mortem lung tissue

Tissue was obtained from a longitudinal clinicopathological donation programme. Donors consented to autopsy and post-mortem tissue donation. Diagnostic examination, including gross and microscopic assessment, was performed by medically licensed pathologists under standardised protocols. SARS-CoV-2 status was determined by licensed laboratories under Emergency Use Authorization protocols.

Multiple histology blocks were taken per donor, sampled from different lobes where available. Each delivered row corresponds to one histology block. Donor-level fields are recorded against the first block for that donor.

Immunohistochemistry was scored semi-quantitatively on a 0–5 scale (0, no expression; 5, high expression) by a single reader. Scores were recorded separately for bronchial and bronchiolar basal epithelium, endothelium, and vascular smooth muscle. Eosinophil, neutrophil and mast cell panels were scored on the same scale on serial or duplex sections as indicated in the column headers.

8. Delivered files

File	Contents
`mmc2.xlsx`	per-assay odds ratios, confidence intervals, p values and adjusted p values against the composite endpoint
`mmc3.xlsx`	per-block donor demographics, pathology text, SARS-CoV-2 status and immunohistochemistry scores
`mmc4.xlsx`	contrast output, IL-33red versus vehicle in HUVEC, both timepoints
`mmc5.xlsx`	contrast output, all remaining contrasts extracted from the fitted model

End of annex.